RACE
Regular Analysis Through Continuous Exercise

Calculation of M.I. : RACE-01

1. Two discs of same mass and different radii are made of different materials such that their thicknesses are 1 *cm* and 0.5 *cm* respectively. The densities of materials are in the ratio 3 : 5. The moment of inertia of these discs respectively about their diameters will be in the ratio of $\frac{x}{6}$. The value of *x* is_____.

PRM322

2. Moment of inertia (M.I.) of four bodies, having same mass and radius, are reported as ;

I_1 = M.I. of thin circular ring about its diameter.

I_2 = M.I. of circular disc about an axis perpendicular to the disc and going through the centre,

I_3 = M.I. of solid cylinder about its axis and

I_4 = M.I. of solid sphere about its diameter.

Then:

(A) $I_1 + I_3 < I_2 + I_4$ (B) $I_1 + I_2 = I_3 + \frac{5}{2} I_4$ (C) $I_1 = I_2 = I_3 > I_4$ (D) $I_1 = I_2 = I_3 < I_4$

PRM323

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3. Consider a uniform wire of mass *M* and length *L*. It is bent into a semicircle. Its moment of inertia about a line perpendicular to the plane of the wire passing through the centre is :

(A) $\frac{1}{4}\frac{ML^2}{\pi^2}$ (B) $\frac{2}{5}\frac{ML^2}{\pi^2}$ (C) $\frac{ML^2}{\pi^2}$ (D) $\frac{1}{2}\frac{ML^2}{\pi^2}$

PRM324

4. Match List-I with List-II:

	List-I		List-II
A	Moment of inertia of solid sphere of radius *R* about any tangent	I	$\frac{5}{3}MR^2$
B	Moment of inertia of hollow sphere of radius (R) about any tangent	II	$\frac{7}{5}MR^2$
C	Moment of inertia of circular ring of radius (R) about its diameter.	III	$\frac{1}{4}MR^2$
D	Moment of inertia of circular disc of radius (R) about any diameter.	IV	$\frac{1}{2}MR^2$

Question: Choose the correct answer from the options given below:

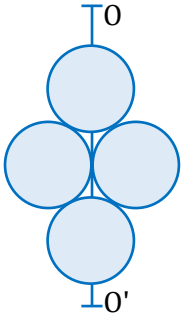
(A) A-II, B-II, C-IV, D-III (B) A-I, B-II, C-IV, D-III

(C) A-II, B-I, C-III, D-IV (D) A-I, B-II, C-III, D-IV

PRM325

For More Join: @IITJEE_Advanced

5. Four identical discs each of mass 'M' and diameter 'a' are arranged in a small plane as shown in figure. If the moment of inertia of the system about OO' is $\frac{x}{4}Ma^2$. Then, the value of x will be ____.



PRM326

6. A uniform thin bar of mass 6 kg and length 2.4 meter is bent to make an equilateral hexagon. The moment of inertia about an axis passing through the centre of mass and perpendicular to the plane of hexagon is _____ $\times 10^{-1}\ kg\ m^2$.

PRM327

7. Four identical solid spheres each of mass 'm' and radius 'a' are placed with their centres on the four corners of a square of side 'b'. The moment of inertia of the system about one side of square where the axis of rotation is parallel to the plane of the square is :

(A) $\frac{4}{5}ma^2+2mb^2$ (B) $\frac{8}{5}ma^2+mb^2$ (C) $\frac{8}{5}ma^2+2mb^2$ (D) $\frac{4}{5}ma^2$

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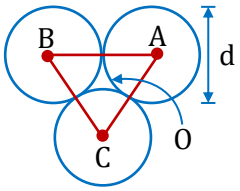
PRM328

8. Moment of inertia of a square plate of side l about the axis passing through one of the corner and perpendicular to the plane of square plate is given by :

(A) $\frac{Ml^2}{6}$ (B) Ml^2 (C) $\frac{Ml^2}{12}$ (D) $\frac{2}{3}Ml^2$

PRM329

9. Three solid spheres each of mass m and diameter d are stuck together such that the lines connecting the centres form an equilateral triangle of side of length d. The ratio I_0/I_A of moment of inertia I_0 of the system about an axis passing the centroid and about center of any of the spheres I_A and perpendicular to the plane of the triangle is :



(A) $\frac{13}{23}$ (B) $\frac{15}{13}$ (C) $\frac{23}{13}$ (D) $\frac{13}{15}$

PRM330

For More Join: @IITJEE_Advanced

10. Moment of inertia of a cylinder of mass M, length L and radius R about an axis passing through its centre and perpendicular to the axis of the cylinder is $I = M\left(\frac{R^2}{4} + \frac{L^2}{12}\right)$. If such a cylinder is to be made for a given mass of material, the ratio L/R for it to have minimum possible I is :-

(A) $\sqrt{\frac{2}{3}}$ (B) $\frac{3}{2}$ (C) $\sqrt{\frac{3}{2}}$ (D) $\frac{2}{3}$

PRM331

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ANSWER KEY

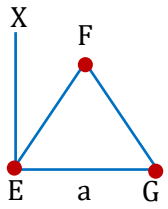
Question	1	2	3	4	5	6	7	8	9	10
Answer	5	C	C	A	3	8	C	D	A	C

For More Join: @IITJEE_Advanced

RACE Regular Analysis Through Continuous Exercise

Moment of Inertia : RACE-02

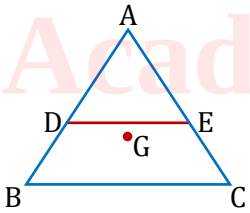
1. An massless equilateral triangle *EFG* of side '*a*' (As shown in figure) has three particles of mass *m* situated at its vertices. The moment of inertia of the system about the line *EX* perpendicular to *EG* in the plane of *EFG* is $\frac{N}{20}ma^2$ where *N* is an integer. The value of *N* is _____.



PRM332

2. *ABC* is a plane lamina of the shape of an equilateral triangle. *D*, *E* are mid points of *AB*, *AC* and *G* is the centroid of the lamina. Moment of inertia of the lamina about an axis passing through *G* and perpendicular to the plane *ABC* is I_0. If part *ADE* is removed, the moment of inertia of the remaining part about the same axis is $\frac{NI_0}{16}$ where *N* is an integer. Value of *N* is ____.

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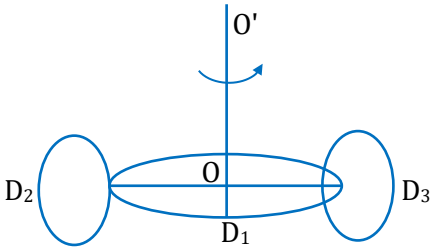
PRM333

3. The linear mass density of a thin rod *AB* of length *L* varies from *A* to *B* as $\lambda(x)=\lambda_0\left(1+\frac{x}{L}\right)$, where *x* is the distance from *A*. If *M* is the mass of the rod then its moment of inertia about an axis passing through *A* and perpendicular to the rod is:

(A) $\frac{5}{12}ML^2$ (B) $\frac{3}{7}ML^2$ (C) $\frac{2}{5}ML^2$ (D) $\frac{7}{18}ML^2$

PRM334

4. A circular disc D_1 of mass *M* and radius *R* has two identical discs D_2 and D_3 of the same mass *M* and radius *R* attached rigidly at its opposite ends (see figure). The moment of inertia of the system about the axis *OO'*, passing through the centre of D_1, as shown in the figure, will be:-



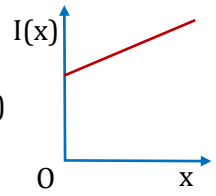
(A) 3 MR^2 (B) $\frac{2}{3}MR^2$

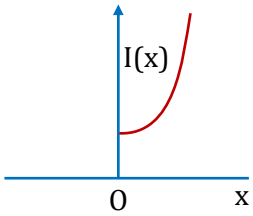
(C) MR^2 (D) $\frac{4}{5}MR^2$

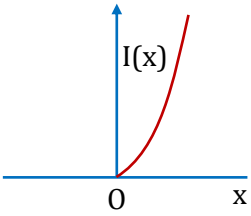
PRM335

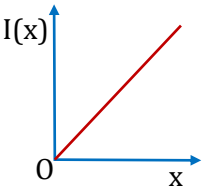
For More Join: @IITJEE_Advanced

5. The moment of inertia of a solid sphere, about an axis parallel to its diameter and at a distance of x from it, is $I(x)$'. Which one of the graphs represents the variation of $I(x)$ with x correctly?

(A) 

(B) 

(C) 

(D) 

PRM336

6. A thin circular plate of mass M and radius R has its density varying as $\rho(r) = \rho_0 r$ with ρ_0 as constant and r is the distance from its centre. The moment of Inertia of the circular plate about an axis perpendicular to the plate and passing through its edge is $I = aMR^2$. The value of the coefficient a is

(A) $\frac{3}{2}$ (B) $\frac{1}{2}$ (C) $\frac{3}{5}$ (D) $\frac{8}{5}$

PRM337

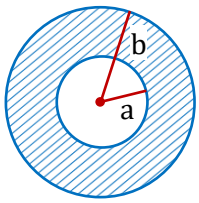
Rankers Academy JEE

7. A solid sphere of mass M and radius R is divided into two unequal parts. The first part has a mass of $\frac{7M}{8}$ and is converted into a uniform disc of radius $2R$. The second part is converted into a uniform solid sphere. Let I_1 be the moment of inertia of the disc about its axis and I_2 be the moment of inertia of the new sphere about its axis. The ratio I_1/I_2 is given by :

(A) 185 (B) 65 (C) 285 (D) 140

PRM338

8. A circular disc of radius b has a hole of radius a at its centre (see figure). If the mass per unit area of the disc varies as $\left(\frac{\sigma_0}{r}\right)$, then the radius of gyration of the disc about its axis passing through the centre is :



(A) $\frac{a+b}{2}$ (B) $\frac{a+b}{3}$ (C) $\sqrt{\frac{a^2+b^2+ab}{2}}$ (D) $\sqrt{\frac{a^2+b^2+ab}{3}}$

PRM339

For More Join: @IITJEE_Advanced

9. A ring and a solid sphere rotating about an axis passing through their centers have same radii of gyration. The axis of rotation is perpendicular to plane of ring. The ratio of radius of ring to that of sphere is $\sqrt{\frac{2}{x}}$. The value of x is _____.

PRM340

10. A solid sphere and a solid cylinder of same mass and radius are rolling on a horizontal surface without slipping. The ratio of their radius of gyrations respectively ($k_{sph} : k_{cyl}$) is $2:\sqrt{x}$, then value of x is _____.

PRM341

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ANSWER KEY

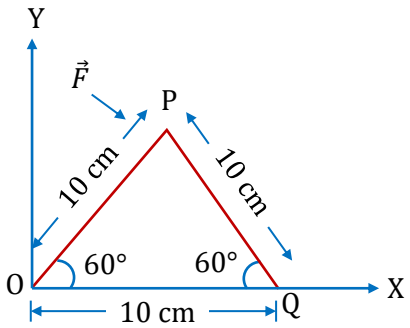
Question	1	2	3	4	5	6	7	8	9	10
Answer	25	11	D	A	B	D	D	D	5	5

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Torque and Angular Acceleration : RACE-03

1. A triangular plate is shown. A force $\vec{F}=4\hat{i}-3\hat{j}$ is applied at point *P*. The torque at point *P* with respect to point '*O*' and '*Q*' are :



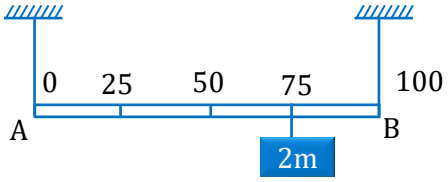
(A) $-15-20\sqrt{3}$, $15-20\sqrt{3}$
(B) $15+20\sqrt{3}$, $15-20\sqrt{3}$
(C) $15-20\sqrt{3}$, $15+20\sqrt{3}$
(D) $-15+20\sqrt{3}$, $15+20\sqrt{3}$

PRM342

2. A solid disc of radius 20 *cm* and mass 10 *kg* is rotating with an angular velocity of 600 *rpm*, about an axis normal to its circular plane and passing through its centre of mass. The retarding torque required to bring the disc at rest in 10*s* is ____ $\pi \times 10^{-1}$ *Nm*.

PRM343

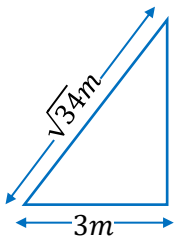
3. Shown in the figure is rigid and uniform one meter long rod *AB* held in horizontal position by two strings tied to its ends and attached to the ceiling. The rod is of mass '*m*' and has another weight of mass 2 *m* hung at a distance of 75 *cm* from *A*. The tension in the string at *A* is :



(A) 2 *mg*
(B) 0.5 *mg*
(C) 0.75 *mg*
(D) 1 *mg*

PRM344

4. A $\sqrt{34}\,m$ long ladder weighing 10 *kg* leans on a frictionless wall. Its feet rest on the floor 3 *m* away from the wall as shown in the figure. If F_f and F_w are the reaction forces of the floor and the wall, then ratio of F_w/F_f will be:
(Use $g = 10\ m/s^2$)

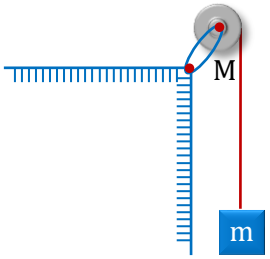


(A) $\frac{6}{\sqrt{110}}$
(B) $\frac{3}{\sqrt{113}}$
(C) $\frac{3}{\sqrt{109}}$
(D) $\frac{2}{\sqrt{109}}$

PRM345

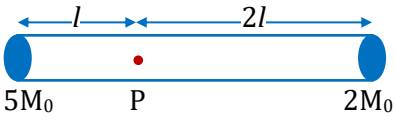
For More Join: @IITJEE_Advanced

5. A uniform disc with mass $M = 4\ kg$ and radius $R = 10\ cm$ is mounted on a fixed horizontal axle as shown in figure. A block with mass $m = 2\ kg$ hangs from a massless cord that is wrapped around the rim of the disc. During the fall of the block, the cord does not slip and there is no friction at the axle. The tension in the cord is _____N. (Take $g = 10\ ms^{-2}$)



PRM346

6. A rigid massless rod of length $3l$ has two masses attached at each end as shown in the figure. The rod is pivoted at point P on the horizontal axis (see figure). When released from initial horizontal position, its instantaneous angular acceleration will be :



(A) $\frac{g}{2l}$ (B) $\frac{7g}{3l}$ (C) $\frac{g}{13l}$ (D) $\frac{g}{3l}$

PRM347

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7. A particle of mass m is moving along a trajectory given by

$x = x_0 + a\cos\omega_1 t$

$y = y_0 + b\sin\omega_2 t$

The torque, acting on the particle about the origin, at $t = 0$ is :

(A) $m\,(-x_0 b + y_0 a)\ \omega_1^2\hat{k}$ (B) $+my_0 a\,\omega_1^2\hat{k}$

(C) $-m(x_0 b\omega_2^2 - y_0 a\omega_1^2)\hat{k}$ (D) Zero

PRM348

8. A pulley of radius 1.5 m is rotated about its axis by a force $F = (12t - 3t^2)\ N$ applied tangentially (while t is measured in seconds). If moment of inertia of the pulley about its axis of rotation is 4.5 $kg\ m^2$, the number of rotations made by the pulley before its direction of motion is reversed, will be $\frac{K}{\pi}$. The value of K is _____.

PRM349

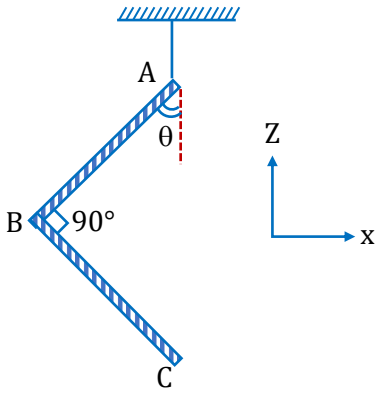
9. The magnitude of torque on a particle of mass $1kg$ is 2.5 Nm about the origin. If the force acting on it is $1N$, and the distance of the particle from the origin is $5m$, the angle between the force and the position vector is (in radians) :-

(A) $\frac{\pi}{8}$ (B) $\frac{\pi}{6}$ (C) $\frac{\pi}{4}$ (D) $\frac{\pi}{3}$

PRM350

For More Join: @IITJEE_Advanced

10. An *L*-shaped object, made of thin rods of uniform mass density, is suspended with a string as shown in figure. If *AB* = *BC*, and the angle made by *AB* with downward vertical is θ, then :



(A) $\tan\theta = \frac{2}{\sqrt{3}}$ (B) $\tan\theta = \frac{1}{3}$ (C) $\tan\theta = \frac{1}{2}$ (D) $\tan\theta = \frac{1}{2\sqrt{3}}$

PRM351

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ANSWER KEY

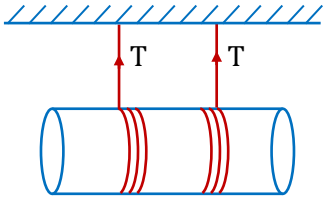
Question	1	2	3	4	5	6	7	8	9	10
Answer	A	4	D	C	10	C	B	18	B	B

For More Join: @IITJEE_Advanced

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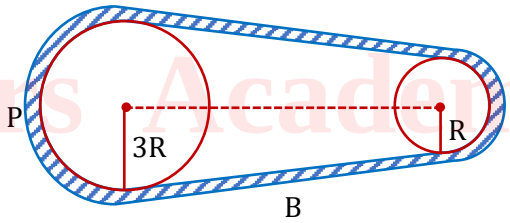
Energy in Rotation : RACE-04

1. A solid cylinder length is suspended symmetrically through two massless strings, as shown in the figure. The distance from the initial rest position, the cylinder should by unbinding the strings to achieve a speed of 4 ms^{-1}, is…….*cm*.
(Take g = 10 ms^{-2})



PRM352

2. In the given figure, two wheels *P* and *Q* are connected by a belt *B*. The radius of *P* is three times as that of *Q*. In case of same rotational kinetic energy, the ratio of rotational inertias $\left(\frac{I_1}{I_2}\right)$ will be x : 1. The value of x will be ____.



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PRM353

3. A thin uniform rod of length 2*m*. cross sectional area '*A*' and density '*d*' is rotated about an axis passing through the centre and perpendicular to its length with angular velocity ω. If value of ω in terms of its rotational kinetic energy E is $\sqrt{\frac{\alpha E}{Ad}}$ then the value of α is _____.

PRM354

4. A disc of mass 1 *kg* and radius *R* is free of rotate about a horizontal axis passing through its centre and perpendicular to the plane of disc. A body of same mass as that of disc is fixed at the highest point of the disc. Now the system is released, when the body comes to the lowest position, its angular speed will be $4\sqrt{\frac{x}{3R}}\ rad\,s^{-1}$ where x = _____. $\left(g = 10 ms^{-2}\right)$

PRM355

5. A cord is wound round the circumference of wheel of radius *r*. The axis of the wheel is horizontal and the moment of inertia about it is *I*. A weight mg is attached to the cord at the end. The weight falls from rest. After falling through a distance '*h*', the square of angular velocity of wheel will be :-

(A) $\frac{2mgh}{I+2mr^2}$ (B) $\frac{2mgh}{I+mr^2}$ (C) 2 gh (D) $\frac{2gh}{I+mr^2}$

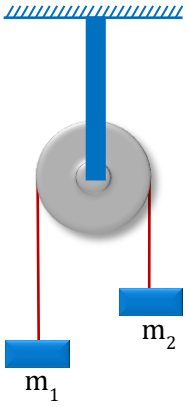
PRM356

For More Join: @IITJEE_Advanced

6. A 2 *kg* steel rod of length 0.6 *m* is clamped on a table vertically at its lower end and is free to rotate in vertical plane. The upper end is pushed so that the rod falls under gravity, Ignoring the friction due to clamping at its lower end, the speed of the free end of rod when it passes through its lowest position is _____ ms^{-1}. (Take $g = 10\ ms^{-2}$)

PRM357

7. A uniformly thick wheel with moment of inertia I and radius R is free to rotate about its centre of mass (see fig). A massless string is wrapped over its rim and two blocks of masses m_1 and m_2 ($m_1 > m_2$) are attached to the ends of the string. The system is released from rest. The angular speed of the wheel when m_1 descents by a distance h is :



(A) $\left[\frac{m_1+m_2}{(m_1+m_2)R^2+I}\right]^{\frac{1}{2}} gh$

(B) $\left[\frac{2(m_1-m_2)gh}{(m_1+m_2)R^2+I}\right]^{\frac{1}{2}}$

(C) $\left[\frac{2(m_1+m_2)gh}{(m_1+m_2)R^2+I}\right]^{\frac{1}{2}}$

(D) $\left[\frac{(m_1-m_2)}{(m_1+m_2)R^2+I}\right]^{\frac{1}{2}} gh$

PRM358

8. A rod of negligible mass and length ℓ is pivoted at its centre. A particle of mass m is fixed to its left end and another particle of mass $2m$ is fixed to the right end. If the system is released from rest and after sometime becomes vertical, the speed v of the two masses and the angular speed at that instant is :-

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(A) $\sqrt{\frac{g\ell}{3}}, \sqrt{\frac{4g}{3\ell}}$

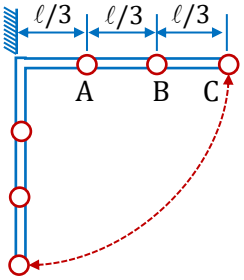
(B) $\sqrt{\frac{4g\ell}{3}}, \sqrt{\frac{4g}{3\ell}}$

(C) $\sqrt{\frac{4g}{3\ell}}, \sqrt{\frac{4g\ell}{3}}$

(D) $\sqrt{\frac{g\ell}{3}}, \sqrt{\frac{g}{3\ell}}$

PRM359

9. A light rod carries three equal masses A, B and C as shown in figure. The velocity of B in vertical position of rod if it is released from horizontal position as shown in figure is :-



(A) $\sqrt{2g\ell}$

(B) $\sqrt{\frac{18g\ell}{7}}$

(C) $\sqrt{\frac{4g\ell}{3}}$

(D) $\sqrt{\frac{8g\ell}{7}}$

PRM360

For More Join: @IITJEE_Advanced

10. A uniform metre stick is held vertically with one end on the floor and is allowed to fall. The speed of the other end when it hits the floor assuming that the end at the floor does not slip :

(A) $\sqrt{4g}$ (B) $\sqrt{3g}$ (C) $\sqrt{5g}$ (D) $\sqrt{g}$

PRM361

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	120	9	3	5	B	6	B	A	D	B

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RACE
Regular Analysis Through Continuous Exercise

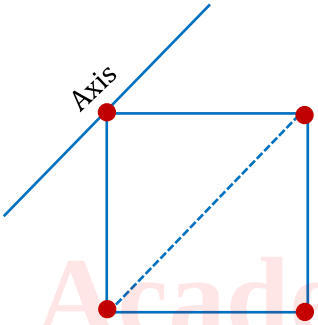
Angular Momentum (Fixed) : RACE-05

1. A particle performs uniform circular motion with an angular momentum L. If the frequency of the particle's motion is tripled and its kinetic energy is halved, the angular momentum becomes :-

(A) $\frac{2L}{3}$ (B) $\frac{3L}{2}$ (C) $\frac{L}{6}$ (D) $6\,L$

PRM362

2. Four point masses, each of mass m, are fixed at the corners of a square of side ℓ. The square is rotating with angular frequency ω, about an axis passing through one of the corners of the square and parallel to its diagonal, as shown in the figure. The angular momentum of the square about this axis is:



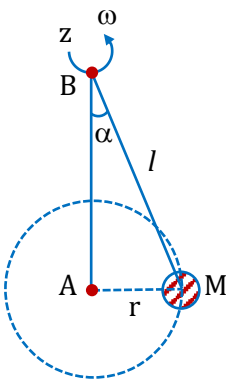
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(A) $2\,m\ell^2\omega$ (B) $3\,m\ell^2\omega$ (C) $m\ell^2\omega$ (D) $4\,m\ell^2\omega$

PRM363

3. A mass M hangs on a massless rod of length l which rotates at a constant angular frequency. The mass M moves with steady speed in a circular path of constant radius. Assume that the system is in steady circular motion with constant angular velocity ω. The angular momentum of M about point A is L_A which lies in the positive z direction and the angular momentum of M about B is L_B. The correct statement for this system is

(A) L_A and L_B are both constant in magnitude and direction.
(B) L_B is constant in direction with varying magnitude.
(C) L_B is constant, both in magnitude and direction.
(D) L_A is constant, both in magnitude and direction.



PRM364

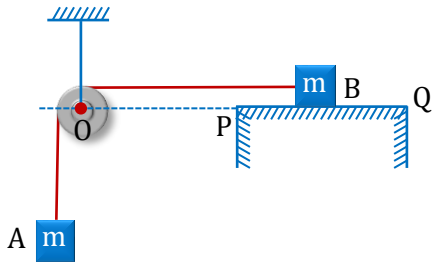
4. Consider a uniform rod of mass $M = 4m$ and length ℓ pivoted about its centre. A mass m moving with velocity v making angle $\theta = \frac{\pi}{4}$ to the rod's long axis collides with one end of the rod and sticks to it. The angular speed of the rod-mass system just after the collision is :

(A) $\frac{3}{7\sqrt{2}}\frac{v}{\ell}$ (B) $\frac{3\sqrt{2}}{7}\frac{v}{\ell}$ (C) $\frac{4}{7}\frac{v}{\ell}$ (D) $\frac{3}{7}\frac{v}{\ell}$

PRM365

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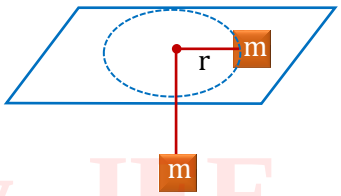
5. In the figure shown, a uniform block *B* is on a horizontal smooth surface *PQ*. *O* is the centre of the pulley (assume it as a uniform disc) of radius *R*. The horizontal part of the string is connected to middle of the left face of the block *B*. String does not slip on the pulley. Points *O*, *P* and *Q* are in the same horizontal line. The masses of *A, B* and the pulley are same and equal to '*m*'. The system is released from rest. Find the angular momentum of the system comprising of *A, B* and the pulley at time '*t*' about '*O*'.



(A) $\frac{mgRt}{2}$ (B) $mgRt$ (C) $2\,mgRt$ (D) $\frac{5}{2}mgRt$

PRM366

6. In the figure the block on the smooth table is set into motion in a circular orbit of radius *r* around the centre hole. The hanging mass is identical to the mass on the table and remains in equilibrium. Neglect friction. The string connecting the two blocks is massless and inextensible. Which is incorrect?



(A) The angular speed ω of the block in its circular motion is $\sqrt{\frac{g}{r}}$.

(B) Kinetic energy as function of *r* is given by $\frac{mgr}{2}$.

(C) Angular momentum about the hole is conserved even if hanging block is pulled down.

(D) The block on table is in equilibrium.

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PRM367

7. A rod of mass *M* and length ℓ stands along *z*-axis. Its lower end is hinged at the centre of a disc of same mass *M* and radius *R*. The whole arrangement is rotating freely about *z*-axis with an angular velocity ω_0. The rod falls on disc and rotates with disc. Find the angular speed of the arrangement.

(A) $\frac{\omega_0 R^2}{\ell^2+R^2}$ (B) $\frac{3\omega_0 R^2}{2\ell^2+3R^2}$ (C) $\frac{6\omega_0 R^2}{\ell^2+6R^2}$ (D) $\frac{6\omega_0 R}{\ell+6R}$

PRM368

8. Two discs have moments of inertia I_1 and I_2 about their respective axes perpendicular to the plane and passing through the centre. They are rotating with angular speeds, ω_1 and ω_2 respectively and are brought into contact face to face with their axes of rotation coaxial. The loss in kinetic energy of the system in the process is given by :

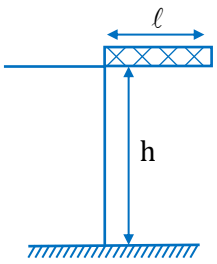
(A) $\frac{I_1 I_2}{(I_1+I_2)}(\omega_1-\omega_2)^2$ (B) $\frac{(I_1-I_2)^2\omega_1\omega_2}{2(I_1+I_2)}$

(C) $\frac{I_1 I_2}{2(I_1+I_2)}(\omega_1-\omega_2)^2$ (D) $\frac{(\omega_1-\omega_2)^2}{2(I_1+I_2)}$

PRM369

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9. A rectangular solid box of length 0.3 m is held horizontally, with one of its sides on the edge of a platform of height 5m. When released, it slips off the table in a very short time $\tau = 0.01s$, remaining essentially horizontal. The angle by which it would rotate when it hits the ground will be (in radians) close to :-



(A) 0.02 (B) 0.28 (C) 0.5 (D) 0.3

PRM370

10. The time dependence of the position of a particle of mass $m = 2$ is given by $\vec{r}(t)=2t\hat{i}-3t^2\hat{j}$. Its angular momentum, with respect to the origin, at time $t = 2$ is :

(A) $36\hat{k}$ (B) $-34(\hat{k}-\hat{i})$ (C) $48(\hat{i}+\hat{j})$ (D) $-48\hat{k}$

PRM371

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ANSWER KEY

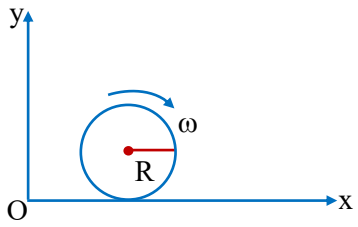
Question	1	2	3	4	5	6	7	8	9	10
Answer	C	B	D	B	B	D	B	C	C	D

For More Join: @IITJEE_Advanced

RACE Regular Analysis Through Continuous Exercise

Translational, Rotational & Angular Momentum Free : RACE-06

1. A spherical shell of 1 *kg* mass and radius *R* is rolling with angular speed ω on horizontal plane (as shown in figure). The magnitude of angular momentum of the shell about the origin *O* is $\frac{a}{3}R^2\omega$. The value of a will be:



(A) 2 (B) 3 (C) 5 (D) 4

PRM372

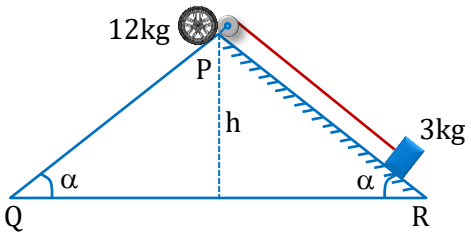
2. A solid cylinder and a solid sphere, having same mass *M* and radius *R*, roll down the same inclined plane from top without slipping. They start from rest. The ratio of velocity of the solid cylinder to that of the solid sphere, with which they reach the ground, will be :

(A) $\sqrt{\frac{5}{3}}$ (B) $\sqrt{\frac{4}{5}}$ (C) $\sqrt{\frac{3}{5}}$ (D) $\sqrt{\frac{14}{15}}$

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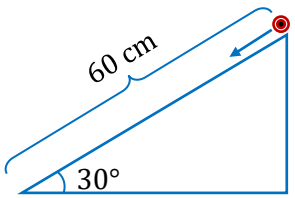
PRM373

3. A rolling wheel of 12 *kg* is on an inclined plane at position *P* and connected to a mass of 3 *kg* through a string of fixed length and pulley as shown in figure. Consider *PR* as friction free surface. The velocity of centre of mass of the wheel when it reaches at the bottom *Q* of the inclined plane *PQ* will be $\sqrt{\frac{x}{3}gh}$ m/s. The value of *x* is _____.



PRM374

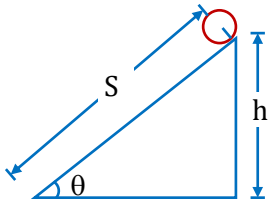
4. A solid cylinder is released from rest from the top of an inclined plane of inclination 30° and length 60 *cm*. If the cylinder rolls without slipping, its speed upon reaching the bottom of the inclined plane is _____ ms^{-1}. (Given $g = 10\ ms^{-2}$)



PRM375

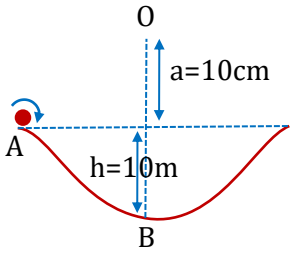
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5. The following bodies,
(1) a ring (2) a disc (3) a solid cylinder (4) a solid sphere,
of same mass 'm' and radius 'R' are allowed to roll down without slipping simultaneously from the top of the inclined plane. The body which will reach first at the bottom of the inclined plane is _____.
[Mark the body as per their respective numbering given in the question]



PRM376

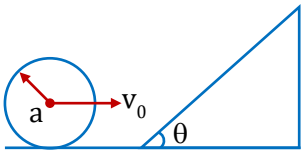
6. A particle of mass 20 g is released with an initial velocity 5 m/s along the curve from the point A, as shown in the figure. The point A is at height h from point B. The particle slides along the frictionless surface. When the particle reaches point B, its angular momentum about O will be :
(Take $g = 10\ m/s^2$)



(A) 8 $kg\text{-}m^2/s$ (B) 6 $kg\text{-}m^2/s$ (C) 3 $kg\text{-}m^2/s$ (D) 2 $kg\text{-}m^2/s$

PRM377

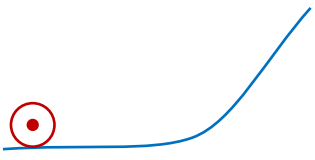
7. A sphere of radius 'a' and mass 'm' rolls along a horizontal plane with constant speed v_0. It encounters an inclined plane at angle θ and climbs upward. Assuming that it rolls without slipping, how far up the sphere will travel ?



(A) $\frac{7v_0^2}{10g\sin\theta}$ (B) $\frac{v_0^2}{5g\sin\theta}$ (C) $\frac{2}{5}\frac{v_0^2}{g\sin\theta}$ (D) $\frac{v_0^2}{2g\sin\theta}$

PRM378

8. A solid sphere and solid cylinder of identical radii approach an incline with the same linear velocity (see figure). Both roll without slipping all throughout. The two climb maximum heights h_{sph} and h_{cyl} on the incline. The ratio $\frac{h_{sph}}{h_{cyl}}$ is given by :-

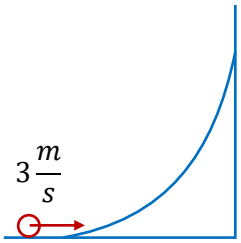


(A) $\frac{14}{15}$ (B) $\frac{4}{5}$ (C) 1 (D) $\frac{2}{\sqrt{5}}$

PRM379

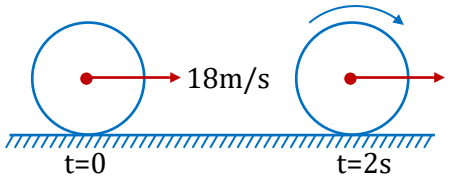
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9. A hollow spherical ball of uniform density rolls up a curved surface with an initial velocity 3 m/s (as shown in figure). Maximum height with respect to the initial position covered by it will be ____ cm.



PRM380

10. A uniform disc of mass 0.5 kg and radius r is projected with velocity 18 m/s at t = 0s on a rough horizontal surface. It starts off with a purely sliding motion at t = 0 s. After 2s it acquires a purely rolling motion (see figure). The total kinetic energy of the disc after 2s will be _____ J.
(Given, coefficient of friction is 0.3 and g = 10 m/s^2).



PRM381

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ANSWER KEY

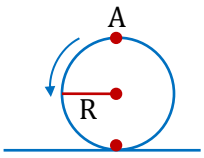
Question	1	2	3	4	5	6	7	8	9	10
Answer	C	D	2	2	4	B	A	A	75	54

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RACE
Regular Analysis Through Continuous Exercise

Rotational Kinematics and Toppling : RACE-07

1. A disc is rolling without slipping on a surface. The radius of the disc is R. At $t = 0$, the top most point on the disc is A as shown in figure. When the disc completes half of its rotation, the displacement of point A from its initial position is



(A) $R\sqrt{(\pi^2+4)}$ (B) $R\sqrt{(\pi^2+1)}$ (C) $2R$ (D) $2R\sqrt{(1+4\pi^2)}$

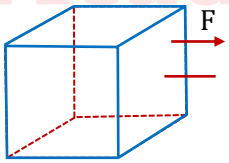
PRM382

2. The centre of a wheel rolling on a plane surface moves with a speed v_0. A particle on the rim of the wheel at the same level as the centre will be moving at a speed $\sqrt{x}\, v_0$. Then the value of x is _____.

PRM383

3. Consider a uniform cubical box of side a on a rough floor that is to be moved by applying minimum possible force F at a point b above its centre of mass (see figure). If the coefficient of friction is $\mu = 0.4$, the maximum possible value of $100 \times \frac{b}{a}$ for a box not to topple before moving is _____.

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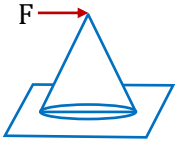
PRM384

4. A homogeneous solid cylindrical roller of radius R and mass M is pulled on a cricket pitch by a horizontal force. Assuming rolling without slipping, angular acceleration of the cylinder is:

(A) $\frac{3F}{2mR}$ (B) $\frac{F}{3mR}$ (C) $\frac{2F}{3mR}$ (D) $\frac{F}{2mR}$

PRM385

5. A cone of radius r and height h rests on a rough horizontal surface, the coefficient of friction between the cone and the surface being μ. A gradually increasing horizontal force F is applied to the vertex of the cone. The largest value of μ for which the cone will slide before it topples is

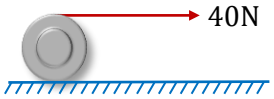


(A) $\mu = \frac{r}{2h}$ (B) $\mu = \frac{2r}{5h}$ (C) $\mu = \frac{r}{h}$ (D) $\mu = \sqrt{\frac{r}{h}}$

PRM386

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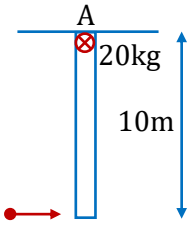
6. A string is wound around a hollow cylinder of mass 5 *kg* and radius 0.5 *m*. If the string is now pulled with a horizontal force of 40 *N*, and the cylinder is rolling without slipping on a horizontal surface (see figure), then the angular acceleration of the cylinder will be
(Neglect the mass and thickness of the string)



(A) 12 rad/s^2 (B) 16 rad/s^2 (C) 10 rad/s^2 (D) 20 rad/s^2

PRM387

7. A rod of mass 20 *kg* and length 10m is hinged at *A* and hanging vertically. A bullet of mass 5 *kg* moving with velocity 10 *m/s* sticks to one end of rod. Find angular velocity of rod just after the collision particle sticks to it :-

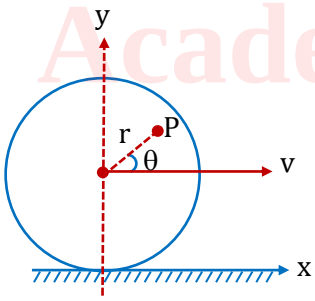


(A) $\frac{7}{3} rad/\sec$ (B) $\frac{3}{7} rad/\sec$ (C) $\frac{3}{10} rad/\sec$ (D) 3 *rad/sec*

PRM388

8. A disc of radius *R* rolls without slipping with speed *v* along positive *x*-axis. Velocity of point *P* at the instant shown in figure is -

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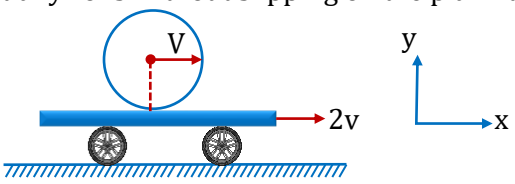


(A) $\vec{V}_p = \left(v + \frac{vr\ \sin\theta}{R}\right)\hat{i} + \frac{vr\ \cos\theta}{R}\hat{j}$ (B) $\vec{V}_p = \left(v + \frac{vr\ \sin\theta}{R}\right)\hat{i} - \frac{vr\ \cos\theta}{R}\hat{j}$

(C) $\vec{V}_p = \frac{vr\ \sin\theta}{R}\hat{i} + \frac{vr\ \cos\theta}{R}\hat{j}$ (D) $\vec{V}_p = \frac{vr\ \sin\theta}{R}\hat{i} - \frac{vr\ \cos\theta}{R}\hat{j}$

PRM389

9. A uniform disc having radius *R* is spin with an angular velocity $\vec{\omega}$ and simultaneously projected with a linear velocity *v* towards right on a plank while the plank moves toward right with a constant velocity 2*v*. If the disc initially rolls without slipping on the plank then angular velocity $\vec{\omega}$ is :-

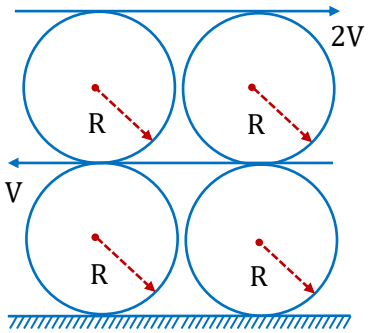


(A) $\frac{v}{R}\hat{k}$ (B) $\frac{v}{R}\left(-\hat{k}\right)$ (C) $\frac{3v}{R}\hat{k}$ (D) $\frac{3v}{R}\left(-\hat{k}\right)$

PRM390

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10. Two thin planks are moving on a four identical cylinder as shown. There is no slipping at any contact points. Calculate the ratio of angular speed of upper cylinder to lower cylinder.



PRM391

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	A	2	75	C	C	B	B	B	A	3

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